

Talk2Sql — SSO Customer Playbook

Reusable reference: what to do when a customer's Oracle Fusion pod uses SSO / SAML federation (Microsoft Entra, Okta, Ping) and you need to run SQL.

This clubs the setup recipe and the troubleshooting journey into one place. Keep it for the next engagement.

0. First question: is the pod SSO federated?

NOT federated (login is a plain Oracle/IDCS username + password)

Just connect with **Basic auth**. Nothing below applies. Done.

Federated (login redirects to Entra/Okta/Ping, often with MFA)

⚠ **This is the problem**. Follow this playbook.

Why SSO is the blocker: a federated user authenticates at the corporate IdP and has **no Fusion local password**. So username/password (Basic) auth to Fusion's data APIs fails, and the REST API / direct report URLs sit behind the SSO web gate. The only programmatic SQL path is: **a local (nonfederated) service account → BI Publisher SOAP → a DBMS_XMLGEN report**. (Same mechanism SplashBI and OFJDBC use.)

1. The fix — 6 steps (you can do all of these yourself with Security Console + BI Author access; no customer dependency)

Step 1 — Create a LOCAL service account

Security Console → Users → Add User Account. Create a **standalone, local** user (e.g. `talk2sql_svc`) — *not* tied to a worker or the Entra directory. **Reset Password** to a known value. A local account has a real Fusion password — the thing federated users lack, and the whole reason this works.

Step 2 — Disable MFA for that account

On the user page → **Configure MFA Exclusion → MFA Excluded → Save**. A headless/service account must never be prompted for MFA.

Step 3 — Assign the STANDARD BI roles

Add `ORA_BI_ADMINISTRATOR_JOB` and `ORA_BI_CONSUMER_JOB`.

A *custom* BI role is **not enough** — the report's permissions key to these standard roles, which also grant the privilege to execute reports. Wait ~3–5 min for role propagation.

Step 4 — Deploy the BI Publisher report (upload + UNARCHIVE)

Download the OFJDBC catalog files from `github.com/krokozyab/ofjdbc` → `otbireport/` :
`DM_ARB.xdm.catalog` (data model) and `RP_ARB.xdo.catalog` (report). In `/xmlpserver` → **Catalog** → **Shared Folders** → **Custom**, create a folder (e.g. `Financials`), then **Unarchive** each `.catalog` into it.

Unarchive, do NOT just upload. A plain upload leaves a `.catalog` file (useless); unarchiving deploys the real **Data Model + Report** objects. After it, you should see a Data Model icon and a Report icon — not `.catalog` files.

Step 5 — Grant the account access to the report

On the folder → **More** → **Permissions** → add the service account with **Read + Execute** on `RP_ARB` and `DM_ARB` (auto-covered if it has `ORA_BI_ADMINISTRATOR_JOB`).

Step 6 — Connect in Talk2Sql

- **Auth:** Basic (the service account from Step 1)
- **Instance URL:** pod base URL, e.g. `https://<pod>.fa.<dc>.oraclecloud.com`
- **Engine:** BIP SQL
- **Report path:** `/Custom/Financials/RP_ARB.xdo`
- Save as a profile.

Smoke test: `SELECT 1 AS ONE FROM DUAL` → 1 row + green "✓ Verified ... → BIP runReport". Then real queries, e.g. `SELECT * FROM CN_COMP_PLANS_ALL_VL FETCH FIRST 5 ROWS ONLY` .

2. What each error means (and the fix)

Error	Meaning	Fix
Redirect to Entra/Okta login page	Hit an SSO-gated endpoint with Basic	Use the SOAP/BIP path + local account
<code>InvalidSecurity ... WS-Security header</code>	Credential can't be validated (federated user, or temp/"must change" password)	Use a local account; clear "must change password" by signing in once locally
<code>does not have permission to run the report</code>	Missing run access	Add <code>ORA_BI_ADMINISTRATOR_JOB</code> + <code>ORA_BI_CONSUMER_JOB</code> ; grant Read/Execute on <code>RP_ARB</code> and <code>DM_ARB</code> ; wait for propagation
<code>Invalid format requested: xml</code>	Report can't emit raw data (interactive <code>xpt</code> layout)	Use the OFJDBC <code>RP_ARB</code> report (DBMS_XMLGEN)
<code>report path not found</code> / 404	Wrong path, or <code>.catalog</code> was uploaded but never unarchived	Unarchive; path is <code>/Custom/Financials/RP_ARB.xdo</code>
<code>ORA-00942: table or view does not exist</code>	Real Oracle error — wrong/invisible table name	Fix the SQL (this means it's executing live — the guard surfaced the real error)

3. Dead ends (don't waste time here on a federated pod)

REST API (`/fscmRestApi`) → 401 · BIP REST API (`/services/rest/v1`) → 404 (not deployed on Fusion Cloud) · SSO session cookie → 401 for REST · `/analytics/saw.dll` → needs a JS-generated token (only a real embedded browser can get it) · direct report URL with Basic → SSO login redirect · your own report with an interactive layout → "Invalid format: xml". **All blocked by design.** The `Step406` path is the one that works.

4. One-line quick reference

SSO pod? Create a **local service account** (no MFA) → give it `ORA_BI_ADMINISTRATOR_JOB` + `ORA_BI_CONSUMER_JOB` → **unarchive** OFJDBC's `DM_ARB` / `RP_ARB` into `/Custom/Financials` → connect Talk2Sql in **Basic + BIP SQL** mode with report path `/Custom/Financials/RP_ARB.xdo`.

5. Security & hygiene

- Read-only by nature (BI layer permits no writes); Talk2Sql also gates to SELECT/WITH only.
- **Rotate the service account password** and store it in a keychain.

- Results carry a live execution provenance stamp; the antifabrication guard means real data or an explicit error — never invented rows.